

Multiplicative Quotient Lifts on Determinant-Two Fibers: Exact Möbius Recovery and Stable Nonpretentiousness

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Abstract

The determinant-two pullback previously represented $\mu(D)\mu(V)$ as a Liouville shift-two product multiplied by two quotient-squarefree masks. We give a different exact representation. For every positive integer c , an explicit one-bounded multiplicative function G_c satisfies

$$G_c(cm) = \mu(m).$$

If $D(z) = d + sz$, $V(z) = u + az$, and $su - ad = 2$, then with $t = aD(z) = ad + asz$,

$$\mu(D(z))\mu(V(z)) = G_a(t)G_s(t+2).$$

Thus the full squarefree information is already inside two multiplicative functions; no prime-square cutoff or CRT squarefree tail is needed for this two-point core.

At prime arguments, G_c differs from the Liouville function only at primes p with $p \parallel c$. Its pretentious distance is therefore bounded below by the Liouville distance minus $2\sum_{p \parallel c} p^{-1}$. For small-polylogarithmic c this loss is lower order than the known logarithmic lower bound. The construction supplies an exact L0 lift and a uniform L1 nonpretentious input. It does not supply physical weights, generic phases, all prefixes, a four-point estimate, positive L2, or a prime-pair theorem.

1 The quotient lift

For a prime p , write $e_p = v_p(c)$. We define the local values of G_c before taking their multiplicative product.

Definition 1.1 (Quotient-Möbius lift). For $c \geq 1$, let $G_c : \mathbb{N} \rightarrow \{-1, 0, 1\}$ be the multiplicative function determined by

$$G_c(p^j) = \begin{cases} (-1)^j, & 0 \leq j < e_p, \\ \mu(p^{j-e_p}), & j \geq e_p. \end{cases} \quad (1)$$

The two cases agree with $G_c(1) = 1$. This is a multiplicative definition through independent prime-power factors. It is generally not completely multiplicative.

Theorem 1.2 (Exact quotient recovery). For every $c, m \geq 1$,

$$\boxed{G_c(cm) = \mu(m)}. \quad (2)$$

Proof. For each prime p , the exponent of p in cm is $e_p + v_p(m)$, which lies in the second case of (1). Its local factor is $\mu(p^{v_p(m)})$. Multiplication over primes gives $\mu(m)$. $\square$

Remark 1.3 (Why the first case is retained). Only values on multiples of c are needed for (2); the values below e_p may therefore be chosen. The choice $(-1)^j$ makes the prime values agree with $\lambda(p) = -1$ unless $e_p = 1$, which minimizes the pretentious-distance perturbation used below.

2 The determinant-two fiber

Let

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2, \quad (3)$$

with positive integral values on the interval under consideration. In the TPC carrier, a, s are positive, odd and coprime [3]. Put

$$Q_{\text{fib}} = as, \quad t(z) = aD(z) = ad + Q_{\text{fib}}z. \quad (4)$$

Theorem 2.1 (Exact determinant-two Möbius lift). *For every integer z on which the forms are positive,*

$$t(z) + 2 = sV(z), \quad (5)$$

$$t(z) \equiv ad \pmod{Q_{\text{fib}}}, \quad (6)$$

$$\mu(D(z))\mu(V(z)) = \boxed{G_a(t(z))G_s(t(z) + 2)} \quad (7)$$

Proof. The determinant equation gives

$$sV(z) - aD(z) = su - ad = 2,$$

which proves (5); (6) follows from (4). Both $a \mid t(z)$ and $s \mid t(z) + 2$. Theorem 1.2 gives

$$G_a(t(z)) = \mu(t(z)/a) = \mu(D(z)), \quad G_s(t(z) + 2) = \mu((t(z) + 2)/s) = \mu(V(z)).$$

□

Corollary 2.2 (No squarefree truncation in the two-point core). *The right side of (7) contains the full Möbius values. Applying a two-point theorem directly to $G_a(t)G_s(t + 2)$ requires neither quotient-squarefree masks nor a prime-square cutoff. Consequently it creates no squarefree truncation tail.*

This is a new representation of the same literal core; it does not delete the remaining coprimality, periodic, physical or prefix factors of the full packet.

3 Pretentious distance under the lift

For one-bounded multiplicative functions, write

$$M(g; Y, Q) = \inf_{\substack{|v| \leq Y \\ q \leq Q \\ \chi \pmod{q}}} \sum_{p \leq Y} \frac{1 - \text{Re}(g(p)\overline{\chi(p)}p^{-iv})}{p}.$$

The convention is the squared pretentious distance used in Tao and Teräväinen [2].

Lemma 3.1 (Prime-level perturbation). *For every prime p ,*

$$G_c(p) = \begin{cases} 1, & p \parallel c, \\ -1, & p \nmid c. \end{cases}$$

Thus $G_c(p) \neq \lambda(p)$ precisely when $p \parallel c$.

Proof. If $v_p(c) = 0$, the second line of (1) gives $\mu(p) = -1$. If $v_p(c) = 1$, it gives $\mu(1) = 1$. If $v_p(c) \geq 2$, the first line gives $(-1)^1 = -1$. □

Proposition 3.2 (Uniform nonpretentious stability). *For all $Y, Q \geq 2$,*

$$\boxed{M(G_c; Y, Q) \geq M(\lambda; Y, Q) - 2 \sum_{\substack{p \leq Y \\ p \parallel c}} \frac{1}{p}} \quad (8)$$

Proof. Fix one χ and v in the defining infimum. By [lemma 3.1](#), the two prime sums agree outside $p \parallel c$. At an exceptional prime, changing one unit-disc value to another changes the real part by at most two. Hence the G_c sum is at least the λ sum minus the displayed error. Taking the infimum proves (8). $\square$

Lemma 3.3 (Small-polylogarithmic perturbations are lower order). *For $c \geq 3$,*

$$\sum_{p|c} \frac{1}{p} \ll 1 + \log_3^+ c, \quad \log_3^+ c := \max\{0, \log \log \log c\}.$$

In particular, if $c \leq (\log X)^A$ for fixed A , then this sum is $o(\log_2 X)$.

Proof. For fixed size of the product of the distinct prime divisors, the reciprocal sum is maximized by the smallest primes. The standard primorial and Mertens estimates then give $\sum_{p|c} p^{-1} \ll 1 + \log_3^+ c$. The stated consequence is immediate. $\square$

Matomäki et al. [[1](#), equation (1.12)] prove, uniformly for characters of modulus at most a fixed small power of $\log X$,

$$M(\lambda; X^2, \log^{1/125} X) \geq \left(\frac{1}{3} - o(1)\right) \log_2 X. \quad (9)$$

Combining the preceding results gives the needed source condition.

Corollary 3.4 (Callable nonpretentious branch). *There is an absolute $\alpha > 0$ such that, for every fixed $A > 0$ and all sufficiently large X depending on A , uniformly for $c \leq (\log X)^A$,*

$$\exp M(G_c; X^2, \log^{1/125} X) \gg (\log X)^\alpha.$$

After decreasing α , the nonpretentious hypothesis of Tao and Teräväinen [[2](#), Theorem 3.1] holds with $\mathcal{L} = (\log X)^\alpha$.

Proof. Equations (8)–(9) and [lemma 3.3](#) give

$$M(G_c; X^2, \log^{1/125} X) \geq (1/3 - o(1)) \log_2 X.$$

Exponentiate and choose any fixed $\alpha < 1/3$, with room for the implicit source constant. $\square$

4 Scope and machine certificate

The deterministic script evaluates (1) exactly, verifies (2) for thousands of integer pairs, checks ordinary multiplicativity on coprime inputs, and verifies (7) on several determinant-two fibers. It records the prime modification set $\{p : p \parallel c\}$. These finite checks are regressions for the algebra; they do not numerically prove (9).

Statement	Level and status
$G_c(cm) = \mu(m)$	Exact L0, PROVED.
Determinant-two identity (7)	Exact L0, PROVED.
Pretentious stability (8)	Unconditional comparison, PROVED.
Small-polylog source condition	Sourced quantitative L1, PROVED.
Bounded periodic reassembly	Available through TPC-147 [4].
Physical multiplier, generic phase, all prefixes	OPEN; not supplied.
Four-point input or positive L2	Not proved.
Fixed X -power, 1/400, prime pairs	Not proved.

The lift therefore removes the squarefree-CRT gate for the two-point Möbius core without claiming that a two-point good-scale theorem is the complete H3 input.

References

- [1] Kaisa Matomäki, Maksym Radziwiłł, and Terence Tao. An Averaged Form of Chowla’s Conjecture. *Algebra & Number Theory*, 9(9):2167–2196, 2015. doi: 10.2140/ant.2015.9.2167.
- [2] Terence Tao and Joni Teräväinen. Quantitative Correlations and Some Problems on Prime Factors of Consecutive Integers, 2026.
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- [4] Liang Wang. Periodic Reassembly Inside a Quantitative Multiplicative Correlation Corridor, 2026. TPC-147 manuscript.